

Technical Writing: Key Concepts

Technical writing is a form of writing that explains **technical or specialized information** in a **clear, direct, and useful** way.

Its main purpose is to help the audience **understand something or perform a task**.

- It focuses on a **specific subject** (software, machines, processes, experiments, etc.).
- It gives **clear explanations, directions, instructions**, often supported by **visuals** like charts, diagrams, and tables.
- It is commonly used in workplaces such as **offices, labs, industries, construction sites, classrooms, and kitchens**.
- The writing always has:
 - a **technical subject**,
 - a **specific audience**, and
 - a **clear purpose** (inform, instruct, guide).

Simple Example

Writing steps for installing an app, explaining a machine's parts, or preparing a lab report are all technical writing.

Characteristics of Technical Writing

Technical writing has several essential features. Each ensures the information is easy to understand and practical to use.

a. Subject

- The subject matter must be **factual, accurate, and technical**.
- Personal opinions or emotional statements should not appear.
- Example: "The system uses a 3-layer architecture" (fact), not "I think this system is great."

b. Audience

- The audience is **who will read, listen to, or view** the document.
- Understanding the audience's expectations helps the writer decide the **style, detail level, and format**.
- Example: Instructions for beginners need simple steps; experts need brief summaries.

c. Organization

- Technical writing follows a **logical, structured format** so the reader can follow easily.
- Common features:
 - short paragraphs
 - clear transitions
 - headings and subheadings
 - numbered lists or bullet points
- Example: A manual may follow the order: Introduction → Steps → Warnings → Troubleshooting.

d. Style

- Style refers to how the writer uses **words, sentences, and patterns** to communicate.
- In technical writing:
 - words must be precise

- sentences should be simple and direct
- unnecessary details must be avoided
- Example: Use “Press the power button” instead of “You might want to consider pressing the power button.”

e. Tone

- Tone means the **overall attitude or impression** the writing gives.
- Technical writing uses a **professional, neutral, and objective** tone.
- It avoids informal language, jokes, or emotional expressions.
- Example: “The device may overheat if used continuously for 3 hours.”

f. Special Features

Technical writing often uses extra tools and techniques that improve clarity.

- **Mechanics of writing:** correct grammar, punctuation, spelling.
- **Graphics:** charts, diagrams, tables, icons, flowcharts. These help the audience understand faster.
- **Technical tools:** writers commonly use software (Word, PowerPoint, design tools) and desktop publishing tools for layout and visual design.
- Example : A process diagram showing how data moves through a system is easier to understand than a long paragraph describing it.

Technical Writing vs. Other Types of Writing

Technical writing is often compared with **Academic writing**, **Imaginative writing**, and **Non-technical (Personal) writing**.

Each type differs in purpose, audience, style, and language.

Technical Writing vs. Academic Writing

Feature	Technical Writing	Academic Writing
Format	Strict, formal, structured	Can be formal but often more flexible
Target Audience	Mixed audience—technical and non-technical readers	Readers within a specific field (students, scholars)
Content	Factual, practical, and straightforward	Focused on a specific academic discipline
Writer’s Knowledge	Writer is usually an expert	Writer is often a learner/researcher
Subject Covered	Scientific, technical, procedural topics	Academic subjects (literature, history, science, etc.)
Purpose	To help someone perform a task or make a decision	To inform, analyze, or explain ideas
Language	No figurative language; very direct	May use descriptive or figurative language when needed

Simple Example

- Technical writing: “*Connect the cable to Port A before turning on the device.*”
- Academic writing: “*This theory explains how digital systems communicate through synchronous protocols.*”

Technical Writing vs. Imaginative Writing

Feature	Technical Writing	Imaginative Writing
Nature	Factual, objective, reality-based	Creative, artistic, based on imagination

Purpose	To inform or instruct; often demands a specific action	To entertain, express emotions, or explore ideas
Writing Style	Clear, concise, plain, direct	Complex, expressive, descriptive
Organization	Modular, predictable, easy to follow	Planned but flexible; flows like a story
Grammar	Must be error-free	Also usually accurate, but more freedom
Register (tone)	Semi-formal to formal	Can be formal or informal depending on story/message
Simple Example		
• Technical writing: manual, user guide, report. • Imaginative writing: stories, poems, novels.		

Technical Writing vs. Non-technical (Personal) Writing

Criteria	Technical Writing	Non-technical (Personal) Writing
Audience	General, mixed, technical or non-technical	Friends, family, close circles
Content	Technical, procedural, standard information	Conversational, personal thoughts
Writing Style	Clear, concise, direct	Simple sentences; may use informal expressions
Grammar	Expected to be error-free	Errors may appear and are acceptable in casual writing
Register (tone)	Formal	Informal
Organization	Highly structured with headings, short paragraphs	Less structured

Audience and Purpose in Technical Writing

An **audience** is the group of people who receive your message, whether through:

- Writing
- Speech
- Performance

In workplace communication, the audience is **the person or people you want to influence or inform**.

Key point: The main goal of communication is to **deliver a message clearly and achieve the intended action**.

Why Understanding Audience Matters

To make communication effective, a writer should consider:

- Are the readers **all alike** or a **mixed group**?
- What do they **already know** about the topic?
- What do they **need to know**?
- What are the **psychological and physical contexts** in which the message will be received?

Principle: Technical writing is **audience-centered**—the audience's needs dictate every writing decision.

Meeting the Audience's Needs

A technical writer's primary duty is to **satisfy the audience's need for information**.

To do this effectively, the writer should:

- Know the audience's **background**
- Use language and style the audience **understands**
- Consider **knowledge level, role, interests, cultural background, and personality**

a. Knowledge Level

- Understand what the audience **already knows**.
- Avoid content that is **too advanced** or too basic.
- Example: For beginners, explain steps clearly; for experts, use brief technical terms.

b. Role

- Role refers to the **audience's job or area of expertise**.
- Include information relevant to their **responsibilities**.
- Example: A manual for engineers includes technical specifications; for end-users, it focuses on operation instructions.

c. Interest

- Know what the audience **cares about**.
- Use **common interests** to engage readers.
- Example: Highlight benefits or features that match the audience's priorities.

d. Cultural Background

- Culture affects **thought patterns and communication styles**.
- Adapt messages so that they are **clear, respectful, and effective** across cultures.
- Example: Avoid idioms or jokes that may confuse or offend international readers.

e. Personality

- Audience members absorb information differently.
 - Some prefer **direct facts**
 - Others prefer **summaries or conclusions first**
- Tailor style to match **audience preferences**.
- Example: Technical manual may have detailed steps for some, but a summary table for others.

A technical writer designs messages to fit the audience's knowledge, role, interest, culture, and personality.

This ensures clarity, usability, and effective action.

Planning Documents Purpose

Purpose is the **specific result or outcome** a writer wants the audience to achieve after reading a document.

In technical writing, the **goal is to inform and persuade the audience** so that they take a specific action.

Key Points

- Technical writing leads the audience to **understand, accept, and act**.
- Effective persuasion depends on **audience analysis** and aligning the message with their **interests and needs**.

Questions a Technical Writer Must Ask

1. What **information** needs to be provided?
2. What should the audience be **persuaded or convinced** of?
3. What is the **desired action** after reading the document?

Example

- Document: Manual for assembling a chair

- Purpose: Make the reader **assemble the chair correctly**
- Action: Follow steps in order to complete assembly

Scope

Scope is the **extent or coverage** of the information in the document—what is included and what is excluded.

Key Points

- Decide **what information is necessary** and what can be left out.
- Include only **relevant details** that help the audience perform the desired action.
- Avoid unnecessary information that may **confuse or distract** the reader.

Example

- Writing directions to make tea:
 - Include: ingredients, steps, boiling water, steeping time
 - Exclude: where to buy tea or personal opinions
- Too little information → reader may fail
- Too much irrelevant information → reader may get distracted

Medium

Medium is the **channel or method** used to deliver the information.

Examples of Medium

- Traditional: letters, memos, meetings, radio, TV
- Modern: email, social media, websites, online documents, apps

Key Considerations for Choosing Medium

1. **Appropriateness:** Will this medium reach the audience effectively?
2. **Cost and Time:** Is it efficient in terms of resources?
3. **Availability:** Can the audience access this medium easily?

Example

- Purpose: Quick instruction to employees → use **email**
- Purpose: Training video for new software → use **video tutorial**

The Writing Process

The writing process refers to the stages a writer follows to produce a complete piece of writing. In technical writing, the three core stages are **planning**, **drafting**, and **copyediting**.

A technical writer should understand three important features of this process:

Writing is recursive

Writing does not move in a straight line. Writers often return to earlier stages to add missing points, correct errors, or improve clarity. Reading the draft again and again naturally leads to changes, making writing a back-and-forth activity.

Writing takes time

Writing involves choosing a topic, researching, drafting, revising, editing, and preparing the final document. Each step takes time, so writers must work patiently.

Writing is different for everyone

Different writers prefer different stages. One may enjoy planning, another may enjoy editing. The writing process varies from person to person.

Stages of Writing Process / Process for Technical Writing

1. Planning

Planning is the first and most important step in the technical writing process. It is the stage where the writer chooses a topic, gathers information, and prepares ideas so the document can later be written clearly and logically.

Planning involves several activities.

1.1 Choosing a Topic

Sometimes the topic is assigned by an employer or teacher. In that case, the writer does not need to think about *what* to write, only *how* to handle the topic.

When the topic is given, the writer should:

- review existing materials (articles, manuals, reports) to find what is already known
- identify gaps
- ask questions about audience, purpose, scope, and medium

Example:

Your boss gives you the topic “Prepare a user guide for the new attendance software.”

You now review the software features, check existing documents, understand the target users (staff), and identify what new information is required.

1.2 Brainstorming

Brainstorming means producing ideas freely **before** writing the document. Writers use this to expand their thinking.

Common techniques:

- **Talking ideas and taking notes:** Saying the ideas aloud and writing key points.
- **Flowcharting:** Drawing diagrams that link main ideas.
- **Outlining:** Listing points in order.
- **Combination of techniques:** Using two or more methods together.

Example:

For writing “How to install WiFi router,” you might list steps, draw a small diagram of connections, and note possible errors users may face.

1.3 Freewriting

Freewriting helps generate ideas without worrying about grammar or structure. It releases mental pressure.

Three types:

- **Open freewriting:** Write whatever comes.
Example: You start writing about “technology in classrooms” and your writing naturally moves to “online teaching tools.”
- **Focused freewriting:** Choose a topic and write only about that subject.
Example: Write freely for 10 minutes only on “Causes of poor user experience in apps.”
- **Looping:** Write continuously for 5 minutes → stop → summarize in one sentence → use that sentence to start the next loop.
Example: After looping you may get a refined sentence like: “The major challenge in app usability is unclear navigation.”

1.4 Journaling

Journaling means keeping a notebook to record ideas, experiences, or reactions. Many writers use journals to gather future topics and examples.

Example:

A technical writer might note:

“Users struggled to find the ‘Export to PDF’ button today.”

This can later become an improved help guide.

1.5 Shaping an Idea

Once ideas are gathered, they must be shaped and organized. This includes questioning, reading, mapping, and outlining.

a. Questioning

Using wh-questions (who, what, when, where, why, how) to obtain missing information.

Example:

Topic: “Online banking security”

Questions:

- Who faces the highest risk?
- What steps reduce attacks?
- How does two-factor authentication work?

b. Mapping

A visual diagram showing parts of the topic and their connection.

Example:

A map for “Solar energy system” may include:

Panels → Inverter → Battery → Wiring → Output

c. Outlining

A structured list of main points in the order they will appear in the document.

Example Outline:

1. Introduction
2. Components
3. Installation steps
4. Safety precautions
5. Troubleshooting

Outlining ensures clarity and flow.

2. Drafting and Revising

Drafting

Drafting is the step where you actually start writing your content based on your planning.

It develops your ideas into **paragraphs, sentences, and a readable structure**.

Key points

- A draft is the *first version* of your writing (not final).
- The goal is **idea development**, not perfection.

Example

- If your topic is “**Impact of AI on Education**”, your draft may look rough:
“AI helps in personalization. Students can get help anytime. But there are limitations too...”

You are simply getting ideas out.

Revising

Revising means improving the draft for:

- clarity
- coherence
- structure
- logical flow
- better vocabulary and tone

It involves **reorganizing content**, adding missing points, removing unnecessary details, and selecting better words.

What we do while revising

- Add notes on the draft
- Shift paragraphs
- Improve transitions
- Correct weak arguments

Example

Draft line → *“AI is good. It helps students.”*

Revised line → *“AI enhances personalized learning by offering tailored feedback to each student.”*

3. Copyediting and Publishing

Copyediting

Copyediting focuses on **correctness**, not ideas.

You check:

- spelling
- grammar
- punctuation
- sentence structure
- consistency in formatting

Tip: Reading aloud helps catch awkward sentences and grammar errors.

Example

Draft: *“AI help student to study better.”*

Copyedited: *“AI helps students study better.”*

Publishing

Publishing means **sending or submitting** the final document to the intended audience.

This could be:

- a teacher
- an employer
- a client
- a team
- a public website

The final document must look professional.

Example

Submitting a polished report to your company about “Network Security Risks”.

4. Writing Collaboratively

Most office and technical writing tasks are done in groups.

Examples:

- newsletters
- project proposals
- brochures
- research reports
- website content

The longer or more complex the task is, the more likely collaboration is required.

Advantages

1. Brings together different knowledge
2. Combines different talents
3. Includes multiple perspectives
4. Improves workplace relationships
5. Often enjoyable
6. Produces stronger final content

Disadvantages

1. Can be frustrating if the team doesn't coordinate
2. Conflicts may arise
3. Can reduce personal motivation
4. Takes longer than working alone
5. Workload may be unequal
6. Final writing may feel disconnected if styles don't match

Brief Correspondence

Communication is a core part of business decision-making.

Although some communication happens face-to-face, most of it takes place through **text messages, memos, e-mails, and letters**.

These are all forms of **brief correspondence**, each with its own advantages.

- **Text messages** → fastest, most efficient
- **E-mails** → formal, quick, widely used
- **Memos & letters** → slower, more structured, often used inside organizations or for official communication

All brief correspondence aims to create **goodwill**—a positive feeling of trust, respect, and cooperation.

Writers must follow the principles of clear and effective communication.

Text Message

Meaning of “Text”

A *text* is any stretch of language—written or spoken—that gives a complete message.

It can be:

- one word: “*PARKING*”
- a short note
- a full book

Every text has identifiable features such as:

- font, alignment, spacing
- bullets and numbering
- grammatical structure
- coherence
- margins, headers, footers

These elements allow us to recognize and analyze a text.

Meaning of “Message”

A message is information passed from one person to another.

It can be:

- **verbal** (spoken, written words)
- **non-verbal** (gesture, posture, expression, wink)

Electronic messages are usually transmitted through teletext or videotext.

Text Message (combined meaning)

The term combines **text + message**, referring to any written/typed message in human history.

It can appear as:

- soft copy (mobile, computer)
- hard copy (paper)
- archived form (saved records)

In business and technical fields, text messages are important for quick, direct communication.

Example

Q: Send a text to your colleague asking for the meeting time.

A:

“Hi Raj, could you please tell me the time for today’s team meeting? Thanks.”

Q: Reply confirming the meeting time.

A:

“Hi Sita, the meeting is scheduled at 3 PM in the conference room. See you there!”

Electronic Mails (E-Mails)

E-mail is the process of sending messages electronically through a computer network.

It allows **fast, low-cost, global communication**.

Messages are mostly encoded in plain ASCII text.

Since the goal is **speed and clarity**, e-mails should be brief, clear, and well-formatted.

Guidelines for Better E-Mail Formatting

- Use simple formatting
- Align everything to the left
- Maintain a formal tone
- Use spell-check and correct grammar
- Keep the content short (preferably under 80 characters per line)
- Avoid ALL CAPS (hard to read and looks rude)
- Provide enough necessary information
- Write a clear and simple **subject line**
- Use a professional e-mail address

Example (good subject line):

“Request for Updated Attendance Report”

Composing an E-Mail

When you click **“Compose New Message”**, you typically see these fields:

To:

The main recipient’s e-mail address

CC:

People who should receive a copy of the message

Subject:

A short line explaining the purpose of the e-mail

Body:

The main message you want to communicate

Q. Compose an email to your immediate boss about the arrangements you have made for the upcoming IT fair at your campus premises.

Answer :

To: [Boss’s email address]

CC: [Optional]

Subject: Arrangements for Upcoming IT Fair

Body:

Dear [Boss’s Name],

I would like to update you on the arrangements I have made for the upcoming IT Fair at our campus:

- **Venue:** Main auditorium and adjoining classrooms
- **Booths:** 12 exhibition booths for student projects

- **Technical Support:** Internet and power connections ready
- **Schedule:** Fair from 10:00 AM to 4:00 PM
- **Volunteers:** 10 student volunteers assigned

Please let me know if any additional arrangements are needed or if changes are required.

Best regards,

[Your Name]

[Your Position]

[Contact Information]

Memo

A **memo** (short for *memorandum*) is a **formal message sent within an organization** to address a specific need.

Memos are one of the most important forms of internal written communication. They are typically **short, official notes** sent to an individual or a group.

Key Components of a Memo

1. **Memo ID Lines** – Includes essential information such as:
 - **From:** Sender's name
 - **To:** Receiver's name
 - **Date:** Date of the memo
 - **Subject/RE:** Topic of the memo
2. **Body of the Memo** – Usually divided into three parts:
 - **Introduction:** State the purpose of the memo
 - **Discussion:** Provide details, explanations, or background
 - **Conclusion:** Give a directive, recommendation, or closing remark

Q. Your company is introducing a new work-from-home policy. Write a memo to inform all employees about this policy.

Answer :

FROM: Sita Sharma, HR Manager, TechSolutions Pvt. Ltd.

TO: All Employees

DATE: 2078-8-20

RE: New Work-from-Home Policy

Body:

Introduction:

This memo is to inform all employees about the new work-from-home (WFH) policy, effective from next Monday.

Discussion:

- Employees can work from home **two days per week**.
- All team members must be available on **Slack and email** during official working hours.
- Managers will schedule **weekly virtual check-ins** to monitor progress.
- Employees are responsible for ensuring a proper home workspace and stable internet connection.

Conclusion:

Please read the policy carefully and follow the guidelines. For any questions, contact the HR department at **hr@techsolutions.com**.

Types of Memo

1. **Inquiry Memo** – Requests specific information
 - *Example:* Request for cost details of office supplies
2. **Response Memo** – Replies to an inquiry
 - *Example:* Response providing cost information

3. **Policy/Directive Memo** – States a policy or procedure for employees to follow
 - *Example:* Coronavirus Pandemic Policy

Other Types of Memos

1. **Status Memo** – Updates about a project or situation
Structure: Purpose statement → Summary → Methods → Results & Discussion → Conclusions → Recommendations
2. **Negative Memo** – Expresses rejection, disagreement, or complaint
Structure: Problem statement → Discussion → Problem leading to decision → Action
3. **Trip Report Memo** – Sent to a supervisor after a business trip
Structure: Purpose → Summary → Discussion → Action
4. **Field/Lab Report Memo** – Reports on inspections or procedures
Structure: Purpose → Summary → Problem → Methods → Results → Conclusions → Recommendations
5. **Personal Memo** – Conveys private information
Structure: Flexible, varies depending on the sender and purpose

Letter Writing

Letter writing is a widely used form of written communication. Letters can be sent via post, fax, e-mail, or other media. They are used in various sectors such as administration, business, and professional communication. The primary purpose of a letter is to communicate, but it can also be used to inform, thank, request, reply, or convey other messages.

Parts of a Letter

Heading

Includes the sender's address and the date. Usually placed at the right top corner, but it can also be on the left.

Inside Address (Superscription)

The address of the receiver, written below the heading on the left-hand side. Common in official or business letters.

Subject Line

States the main message of the letter. Optional in personal letters.

Salutation

Greeting used to address the recipient. Examples:

- **Sir/Madam:** For strangers or high-ranking officials
- **Dear Sir/Madam:** For business colleagues
- **Dear Mr.:** For familiar recipients
- **My Dear Mr.:** For very close or personal recipients

Body

Contains the main message the sender wants to communicate. Determines the size and quality of the letter.

Subscription (Complimentary Close)

Examples: **Yours truly, Yours sincerely, Yours faithfully, Yours obediently**

Signature

Full name and signature of the sender below the subscription.

Enclosure (if any)

Includes necessary documents mentioned in the letter. Example: Bill, certificate, or other important papers.

Q. Write a letter to the Chief District Officer requesting permission to collect donations for providing about a dozen computers to your former school, including specifications and price list.

Answer :

Heading:

16 Nov, 2081

Kathmandu, Nepal

Inside Address:

Chief District Officer

District Administration Office

Baglung, Nepal

Subject: Request for Permission to Collect Donations for Computers

Salutation:

Dear Sir/Madam,

Body:

I am a current IT student at [Your Campus Name], Tribhuvan University. I wish to donate approximately twelve desktop computers to my former school, [School Name], which currently has very limited computer facilities.

To raise funds for this donation, I kindly request your permission to collect donations from the public. The computers I plan to donate will have the following specifications:

- CPU: Intel Core i5
- RAM: 8 GB
- Storage: 512 GB SSD
- Monitor: 21-inch LED
- Price per unit: NRs. 65,000

I would be grateful for your approval so that this initiative can support the students' learning in my former school.

Thank you for your kind consideration.

Subscription:

Yours faithfully,

Signature:

Aashish Panta

Enclosure:

Computer specifications and price list

16 Nov, 2081

Kathmandu, Nepal

To:

Chief District Officer

District Administration Office

Baglung, Nepal

Subject: Request for Permission to Collect Donations for Computers

Dear Sir/Madam,

I am currently studying IT at [Your Campus Name], Tribhuvan University. I wish to donate about twelve desktop computers to my former school, [School Name], which currently has very limited computer facilities.

To raise funds for this donation, I kindly request your permission to collect donations from the public. The computers I plan to donate will have the following specifications:

- CPU: Intel Core i5
- RAM: 8 GB
- Storage: 512 GB SSD
- Monitor: 21-inch LED
- Price per unit: NRs. 65,000

I would be grateful for your approval so that this initiative can support the students' learning in my former school.

Thank you for your kind consideration.

Yours faithfully,

Aashish Pantia

Enclosure: Computer specifications and price list

Business Letters

A business letter is written to exchange information between individuals and organizations. Its goals are to maintain contact, foster good relationships with the public, and leave a positive impression on customers, employees, and other stakeholders.

Types of Business Letters

- **Enquiry letter:** To ask about quality, price, cost, discount, etc.
- **Quotation letter:** To reply to an enquiry.
- **Order letter:** For detailed description of goods or services purchased.
- **Confirmation letter:** Written by the seller/exporter to confirm the order.
- **Complaint letter:** To complain about goods or services, e.g., low quality.

Audience

All readers share basic characteristics, but the form of communication should remain brief and professional.

- **Text messages:** Audience may be internal or external; the writer may know little or a lot about them. Anticipate their expectations to set the right level of formality.
- **Emails:** Most convenient for internal or external communication. Readers expect clarity and relevance.
- **Memos:** Used internally; audience may have different roles. Writer should gather sufficient information about the reader.
- **Letters:** Represent the writer professionally; must be clear, concise, and complete.

Prewriting

Before writing, analyze your audience and plan carefully:

- What action do I want the receiver to take?
- What is the main point?
- Does the reader need background information?
- Do I need to simplify the message for the audience?
- What questions should my correspondence answer?

Also, gather facts and details like background, events, problems, names, titles, addresses, and check for accuracy.

Composing the Message

The body of the correspondence is the core. Organize ideas to suit the reader, not the writer. Some messages should be straightforward, others more tactful to avoid offending the reader.